Data Analysis
AMS315

Chapter 14

Introduction to Multiple Regression

Learning Objectives

In this chapter, you learn:

- How to develop a multiple regression model
- How to interpret the regression coefficients
- How to determine which independent variables to include in the regression model
- How to determine which independent variables are more important in predicting a dependent variable
- How to use categorical variables in a regression model
- How to predict a categorical dependent variable using logistic regression

The Multiple Regression Model

Idea: Examine the linear relationship between
1 dependent (Y) & 2 or more independent variables (X_i)

Multiple Regression Model with k Independent Variables:

Y-intercept

Population slopes

Random Error

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \ldots + \beta_k X_{ki} + \varepsilon_i$$

Multiple Regression Equation

The coefficients of the multiple regression model are estimated using sample data

Multiple regression equation with k independent variables:

Estimated (or predicted) value of Y

Estimated intercept

Estimated slope coefficients

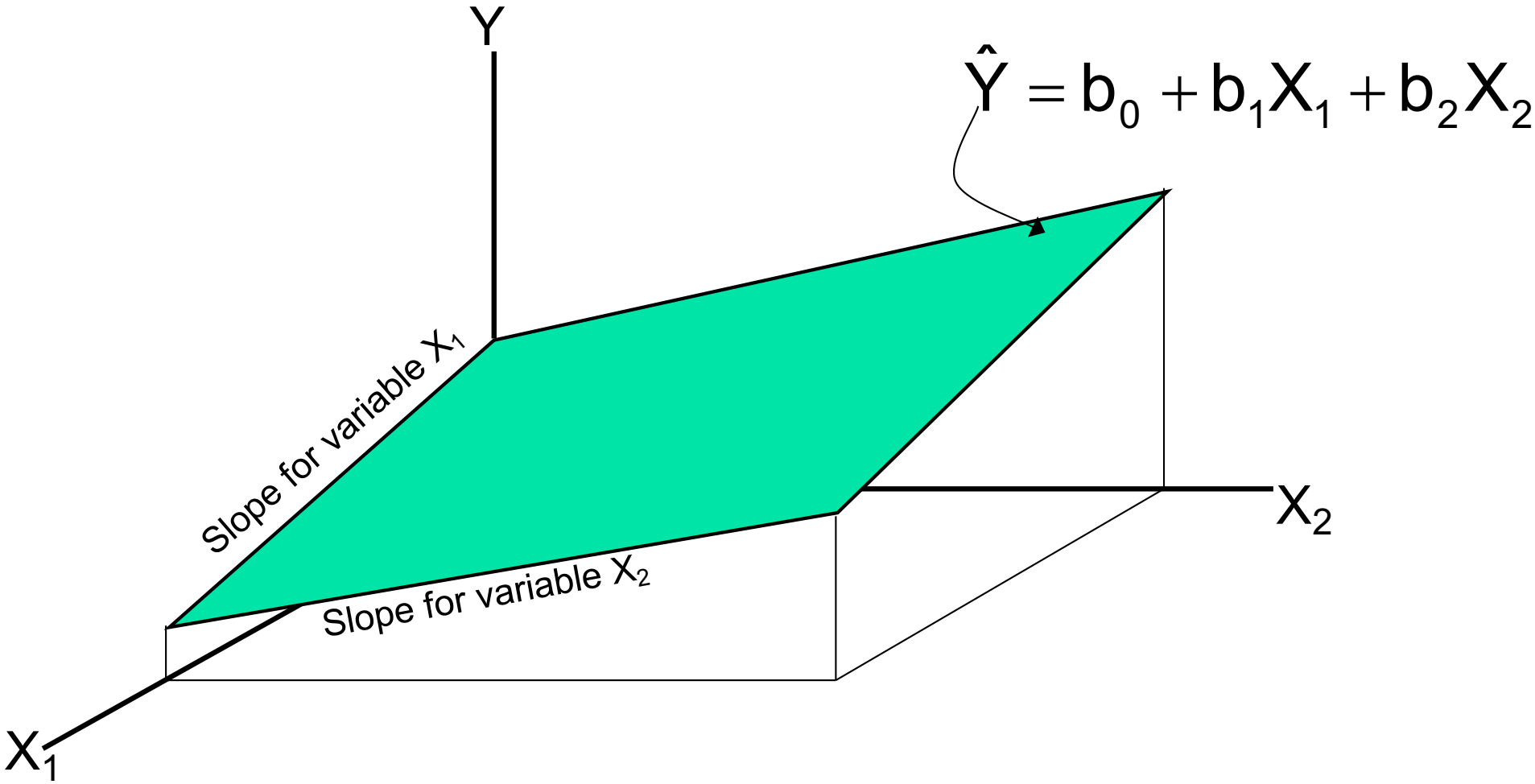
$$\hat{Y}_i = b_0 + b_1 X_{1i} + b_2 X_{2i} + \ldots + b_k X_{ki}$$

In this chapter we will always use Excel to obtain the regression slope coefficients and other regression summary measures.

Multiple Regression Equation

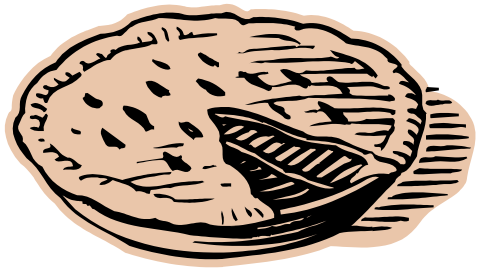
(continued)

Two variable model



Example: 2 Independent Variables

- A distributor of frozen desert pies wants to evaluate factors thought to influence demand
  - Dependent variable: Pie sales (units per week)
  - Independent variables: Price (in $), Advertising ($100's)
- Data are collected for 15 weeks



Pie Sales Example

Week	Pie Sales	Price ($)	Advertising ($100s)
1	350	5.50	3.3
2	460	7.50	3.3
3	350	8.00	3.0
4	430	8.00	4.5
5	350	6.80	3.0
6	380	7.50	4.0
7	430	4.50	3.0
8	470	6.40	3.7
9	450	7.00	3.5
10	490	5.00	4.0
11	340	7.20	3.5
12	300	7.90	3.2
13	440	5.90	4.0
14	450	5.00	3.5
15	300	7.00	2.7

Multiple regression equation:

$$\widehat{\text{Sales}} = b_0 + b_1 (\text{Price}) + b_2 (\text{Advertising})$$

Multiple Regression Output

Regression Statistics	
Multiple R	0.72213
R Square	0.52148
Adjusted R Square	0.44172
Standard Error	47.46341
Observations	15

$$\text{Sales} = 306.526 - 24.975(\text{Price}) + 74.131(\text{Advertising})$$

ANOVA	*df*	*SS*	*MS*	*F*	*Significance F*
Regression	2	29460.027	14730.013	6.53861	0.01201
Residual	12	27033.306	2252.776		
Total	14	56493.333			

	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*
Intercept	306.52619	114.25389	2.68285	0.01993	57.58835	555.46404
Price	-24.97509	10.83213	-2.30565	0.03979	-48.57626	-1.37392
Advertising	74.13096	25.96732	2.85478	0.01449	17.55303	130.70888

The Multiple Regression Equation

$$\widehat{\text{Sales}} = 306.526 - 24.975(\text{Price}) + 74.131(\text{Advertising})$$

where
Sales is in number of pies per week
Price is in $
Advertising is in $100's.

b_1 = -24.975: sales will decrease, on average, by 24.975 pies per week for each $1 increase in selling price, net of the effects of changes due to advertising

b_2 = 74.131: sales will increase, on average, by 74.131 pies per week for each $100 increase in advertising, net of the effects of changes due to price

Using The Equation to Make Predictions

Predict sales for a week in which the selling price is $5.50 and advertising is $350:

$$\widehat{\text{Sales}} = 306.526 - 24.975(\text{Price}) + 74.131(\text{Advertising})$$

$$= 306.526 - 24.975\,(5.50) + 74.131\,(3.5)$$

$$= 428.62$$

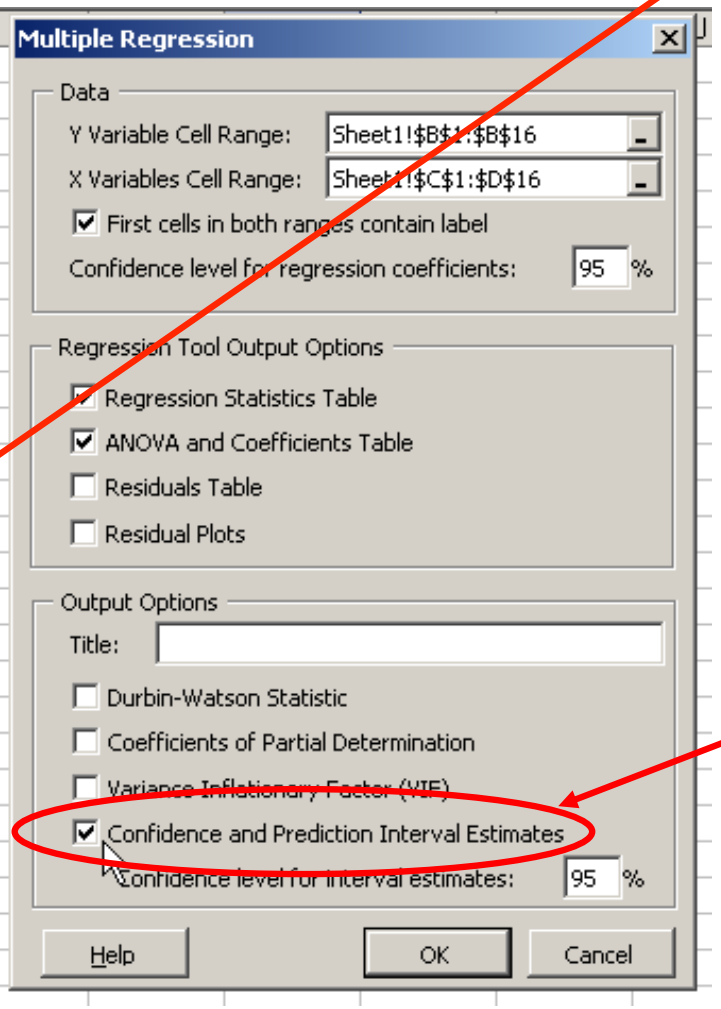
Predicted sales is 428.62 pies

Note that Advertising is in $100's, so $350 means that $X_2 = 3.5$

Predictions in Excel using PHStat

- PHStat | regression | multiple regression …

	A	B	C	D
1	Week	Pie Sales	Price	Advertising
2	1	350	5.5	3.3
3	2	460	7.5	3.3
4	3	350	8	3
5	4	430	8	4.5
6	5	350	6.8	3
7	6	380	7.5	4
8	7	430	4.5	3
9	8	470	6.4	3.7
10	9	450	7	3.5
11	10	490	5	4
12	11	340	7.2	3.5
13	12	300	7.9	3.2
14	13	440	5.9	4
15	14	450	5	3.5
16	15	300	7	2.7



Check the "confidence and prediction interval estimates" box

Predictions in PHStat

(continued)

	A	B
1	Confidence and Prediction Estimate Intervals	
2		
3	Data	
4	Confidence Level	95%
5		
6	Price given value	5.5
7	Advertising given value	3.5
8		
20	t Statistic	2.178813
21	Predicted Y (YHat)	428.6216
22		
23	For Average Predicted Y (Yhat)	
24	Interval Half Width	37.50306
25	Confidence Interval Lower Limit	391.1185
26	Confidence Interval Upper Limit	466.1246
27		
28	For Individual Response Y	
29	Interval Half Width	110.0041
30	Prediction Interval Lower Limit	318.6174
31	Prediction Interval Upper Limit	538.6257

Input values

Predicted $\hat{Y}$ value

Confidence interval for the mean $\hat{Y}$ value, given these X's

Prediction interval for an individual $\hat{Y}$ value, given these X's

Coefficient of Multiple Determination

- Reports the proportion of total variation in Y explained by all X variables taken together

$$r^2 = \frac{SSR}{SST} = \frac{\text{regression sum of squares}}{\text{total sum of squares}}$$

Multiple Coefficient of Determination

(continued)

Regression Statistics	
Multiple R	0.72213
R Square	0.52148
Adjusted R Square	0.44172
Standard Error	47.46341
Observations	15

$$r^2 = \frac{SSR}{SST} = \frac{29460.0}{56493.3} = .52148$$

52.1% of the variation in pie sales is explained by the variation in price and advertising

ANOVA	*df*	*SS*	*MS*	*F*	*Significance F*
Regression	2	29460.027	14730.013	6.53861	0.01201
Residual	12	27033.306	2252.776		
Total	14	56493.333			

	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*
Intercept	306.52619	114.25389	2.68285	0.01993	57.58835	555.46404
Price	-24.97509	10.83213	-2.30565	0.03979	-48.57626	-1.37392
Advertising	74.13096	25.96732	2.85478	0.01449	17.55303	130.70888

Adjusted r^2

- r^2 never decreases when a new X variable is added to the model
  - This can be a disadvantage when comparing models
- What is the net effect of adding a new variable?
  - We lose a degree of freedom when a new X variable is added
  - Did the new X variable add enough explanatory power to offset the loss of one degree of freedom?

Adjusted r^2

(continued)

- Shows the proportion of variation in Y explained by all X variables adjusted for the number of X variables used

$$r_{adj}^2 = 1 - \left[(1 - r^2) \left(\frac{n-1}{n-k-1} \right) \right]$$

(where n = sample size, k = number of independent variables)

- Penalize excessive use of unimportant independent variables
- Smaller than r^2
- Useful in comparing among models

Adjusted r^2

(continued)

Regression Statistics	
Multiple R	0.72213
R Square	0.52148
Adjusted R Square	0.44172
Standard Error	47.46341
Observations	15

$$r^2_{adj} = .44172$$

44.2% of the variation in pie sales is explained by the variation in price and advertising, taking into account the sample size and number of independent variables

ANOVA	*df*	*SS*	*MS*	*F*	*Significance F*
Regression	2	29460.027	14730.013	6.53861	0.01201
Residual	12	27033.306	2252.776		
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Is the Model Significant?

- F Test for Overall Significance of the Model
- Shows if there is a linear relationship between all of the X variables considered together and Y
- Use F-test statistic
- Hypotheses:

H_0: $\beta_1 = \beta_2 = \ldots = \beta_k = 0$ (no linear relationship)

H_1: at least one $\beta_i \neq 0$ (at least one independent variable affects Y)

F Test for Overall Significance

- Test statistic:

$$F = \frac{MSR}{MSE} = \frac{\dfrac{SSR}{k}}{\dfrac{SSE}{n-k-1}}$$

where F has (numerator) = k and

(denominator) = (n – k - 1)

degrees of freedom

F Test for Overall Significance

(continued)

Regression Statistics	
Multiple R	0.72213
R Square	0.52148
Adjusted R Square	0.44172
Standard Error	47.46341
Observations	15

$$F = \frac{MSR}{MSE} = \frac{14730.0}{2252.8} = 6.5386$$

With 2 and 12 degrees of freedom

P-value for the F Test

ANOVA	*df*	*SS*	*MS*	*F*	*Significance F*
Regression	2	29460.027	14730.013	6.53861	0.01201
Residual	12	27033.306	2252.776		
Total	14	56493.333			

	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*
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F Test for Overall Significance

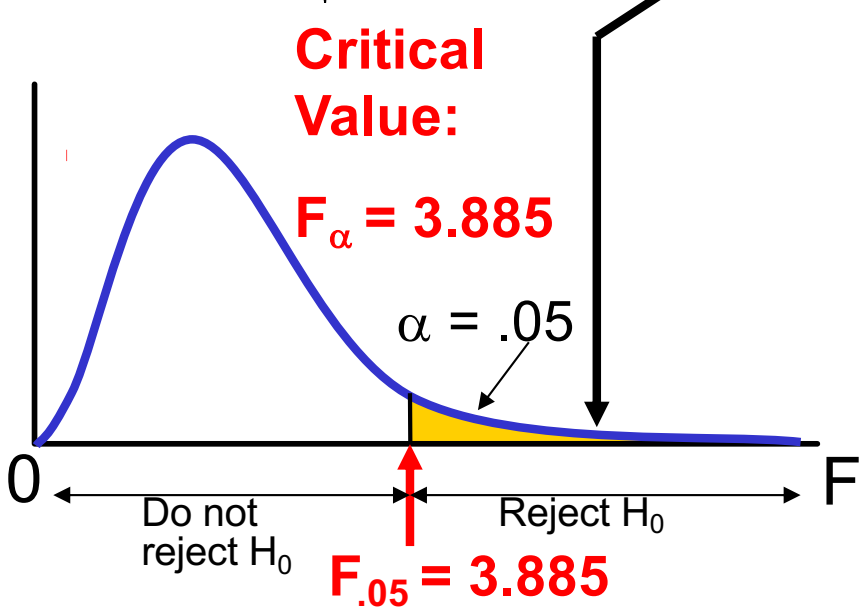
(continued)

H_0: $\beta_1 = \beta_2 = 0$

H_1: β_1 and β_2 not both zero

$\alpha = .05$

$df_1 = 2 \qquad df_2 = 12$



Test Statistic:

$$F = \frac{MSR}{MSE} = 6.5386$$

Decision:

Since F test statistic is in the rejection region (p-value < .05), reject H_0

Conclusion:

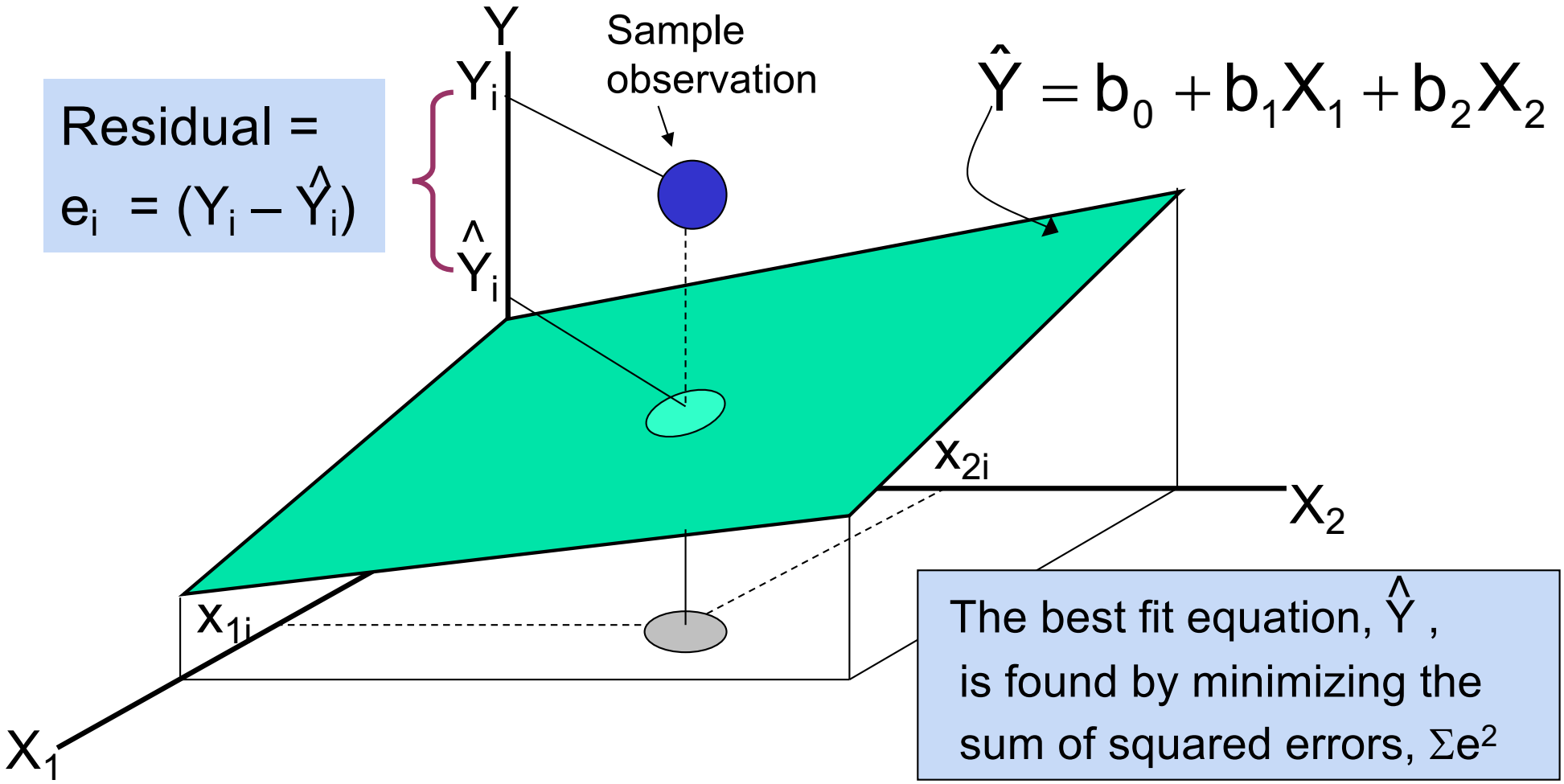
There is evidence that at least one independent variable affects Y

Residuals in Multiple Regression

Two variable model

Residual =

$e_i = (Y_i - \hat{Y}_i)$



The best fit equation, $\hat{Y}$, is found by minimizing the sum of squared errors, Σe^2

Multiple Regression Assumptions

Errors (residuals) from the regression model:

$$e_i = (Y_i - \hat{Y}_i)$$

Assumptions:

- The errors are normally distributed
- Errors have a constant variance
- The model errors are independent

Residual Plots Used in Multiple Regression

- These residual plots are used in multiple regression:
  - Residuals vs. $\hat{Y}_i$
  - Residuals vs. X_{1i}
  - Residuals vs. X_{2i}
  - Residuals vs. time (if time series data)

Use the residual plots to check for violations of regression assumptions

Are Individual Variables Significant?

- Use t tests of individual variable slopes
- Shows if there is a linear relationship between the variable X_j and Y
- Hypotheses:
  - H_0: $\beta_j = 0$ (no linear relationship)
  - H_1: $\beta_j \neq 0$ (linear relationship does exist between X_j and Y)

Are Individual Variables Significant?

(continued)

H_0: $\beta_j = 0$ (no linear relationship)

H_1: $\beta_j \neq 0$ (linear relationship does exist between x_j and y)

Test Statistic:

$$t = \frac{b_j - 0}{S_{b_j}}$$

(df = n – k – 1)

Are Individual Variables Significant?

(continued)

Regression Statistics	
Multiple R	0.72213
R Square	0.52148
Adjusted R Square	0.44172
Standard Error	47.46341
Observations	15

t-value for Price is t = -2.306, with p-value .0398

t-value for Advertising is t = 2.855, with p-value .0145

ANOVA	*df*	*SS*	*MS*	*F*	*Significance F*
Regression	2	29460.027	14730.013	6.53861	0.01201
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Intercept	306.52619	114.25389	2.68285	0.01993	57.58835	555.46404
Price	-24.97509	10.83213	-2.30565	0.03979	-48.57626	-1.37392
Advertising	74.13096	25.96732	2.85478	0.01449	17.55303	130.70888

Inferences about the Slope: t Test Example

$H_0: \beta_i = 0$

$H_1: \beta_i \neq 0$

d.f. = 15-2-1 = 12

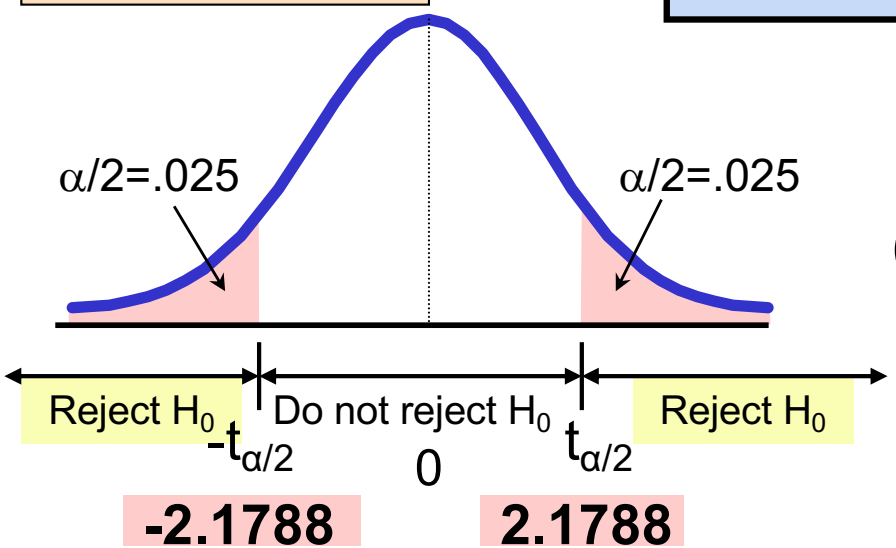
α = .05

$t_{\alpha/2}$ = 2.1788

From Excel output:

	Coefficients	*Standard Error*	*t Stat*	*P-value*
Price	**-24.97509**	**10.83213**	**-2.30565**	**0.03979**
Advertising	**74.13096**	**25.96732**	**2.85478**	**0.01449**

The test statistic for each variable falls in the rejection region (p-values < .05)



Decision:

Reject H_0 for each variable

Conclusion:

There is evidence that both Price and Advertising affect pie sales at α = .05

Confidence Interval Estimate for the Slope

Confidence interval for the population slope β_j

$$b_j \pm t_{n-k-1} S_{b_j}$$

where t has (n – k – 1) d.f.

	Coefficients	*Standard Error*
Intercept	**306.52619**	**114.25389**
Price	**-24.97509**	**10.83213**
Advertising	**74.13096**	**25.96732**

Here, t has (15 – 2 – 1) = 12 d.f.

Example: Form a 95% confidence interval for the effect of changes in price (X_1) on pie sales:

$$-24.975 \pm (2.1788)(10.832)$$

So the interval is (-48.576 , -1.374)

(This interval does not contain zero, so price has a significant effect on sales)

Confidence Interval Estimate for the Slope

(continued)

Confidence interval for the population slope β_i

	Coefficients	*Standard Error*	…	*Lower 95%*	*Upper 95%*
Intercept	**306.52619**	**114.25389**	**…**	**57.58835**	**555.46404**
Price	**-24.97509**	**10.83213**	**…**	**-48.57626**	**-1.37392**
Advertising	**74.13096**	**25.96732**	**…**	**17.55303**	**130.70888**

Example: Excel output also reports these interval endpoints:

Weekly sales are estimated to be reduced by between 1.37 to 48.58 pies for each increase of $1 in the selling price

Testing Portions of the Multiple Regression Model

- Contribution of a Single Independent Variable X_j

$$SSR(X_j \mid \text{all variables except } X_j)$$
$$= SSR(\text{all variables}) - SSR(\text{all variables except } X_j)$$

  - Measures the contribution of X_j in explaining the total variation in Y (SST)

Testing Portions of the Multiple Regression Model

(continued)

Contribution of a Single Independent Variable X_j, assuming all other variables are already included (consider here a 3-variable model):

$$SSR(X_1 \mid X_2 \text{ and } X_3) = SSR(\text{all variables}) - SSR(X_2 \text{ and } X_3)$$

From ANOVA section of regression for

$$\hat{Y} = b_0 + b_1X_1 + b_2X_2 + b_3X_3$$

From ANOVA section of regression for

$$\hat{Y} = b_0 + b_2X_2 + b_3X_3$$

Measures the contribution of X_1 in explaining SST

The Partial F-Test Statistic

- Consider the hypothesis test:

H_0: variable X_j does not significantly improve the model after all other variables are included

H_1: variable X_j significantly improves the model after all other variables are included

- Test using the F-test statistic:

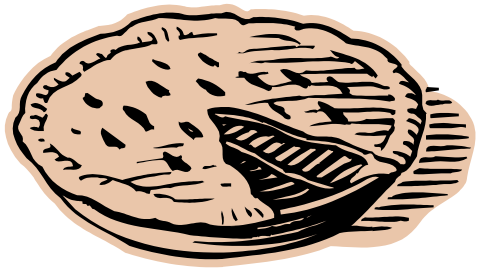
(with 1 and n-k-1 d.f.)

$$F = \frac{SSR(X_j \mid \text{all variables except } j)}{MSE}$$

Testing Portions of Model: Example

Example: Frozen desert pies

Test at the α = .05 level to determine whether the price variable significantly improves the model given that advertising is included



Testing Portions of Model: Example

(continued)

H_0: X_1 (price) does not improve the model with X_2 (advertising) included

H_1: X_1 does improve model

α = .05, df = 1 and 12

F critical Value = 4.75

(For X_1 and X_2)

ANOVA			
	df	*SS*	*MS*
Regression	2	**29460.02687**	14730.01343
Residual	12	27033.30647	**2252.775539**
Total	14	56493.33333	

(For X_2 only)

ANOVA		
	df	*SS*
Regression	1	**17484.22249**
Residual	13	39009.11085
Total	14	56493.33333

Testing Portions of Model: Example

(continued)

(For X_1 and X_2)

ANOVA			
	df	*SS*	*MS*
Regression	2	**29460.02687**	14730.01343
Residual	12	27033.30647	**2252.775539**
Total	14	56493.33333	

(For X_2 only)

ANOVA		
	df	*SS*
Regression	1	**17484.22249**
Residual	13	39009.11085
Total	14	56493.33333

$$F = \frac{SSR(X_1 \mid X_2)}{MSE(all)} = \frac{29{,}460.03 - 17{,}484.22}{2252.78} = 5.316$$

Conclusion: Reject H_0; adding X_1 does improve model

Coefficient of Partial Determination for k variable model

$$r^2_{Yj.(\text{all variables except } j)} = \frac{SSR\,(X_j \mid \text{all variables except } j)}{SST - SSR(\text{all variables}) + SSR(X_j \mid \text{all variables except } j)}$$

- Measures the proportion of variation in the dependent variable that is explained by X_j while controlling for (holding constant) the other explanatory variables

Coefficient of Partial Determination in Excel

- Coefficients of Partial Determination can be found using Excel:
  - PHStat | regression | multiple regression …
    - Check the “coefficient of partial determination” box

Regression Analysis
Coefficients of Partial Determination

Intermediate Calculations			
SSR(X1,X2)	29460.02687		
SST	56493.33333		
SSR(X2)	17484.22249	SSR(X1 | X2)	11975.80438
SSR(X1)	11100.43803	SSR(X2 | X1)	18359.58884

Coefficients	
r2 Y1.2	**0.307000188**
r2 Y2.1	**0.404459524**

Using Dummy Variables

- A dummy variable is a categorical explanatory variable with two levels:
  - yes or no, on or off, male or female
  - coded as 0 or 1
- Regression intercepts are different if the variable is significant
- Assumes equal slopes for other variables
- If more than two levels, the number of dummy variables needed is (number of levels - 1)

Dummy-Variable Example (with 2 Levels)

$$\hat{Y} = b_0 + b_1 X_1 + b_2 X_2$$

Let:

Y = pie sales

X_1 = price

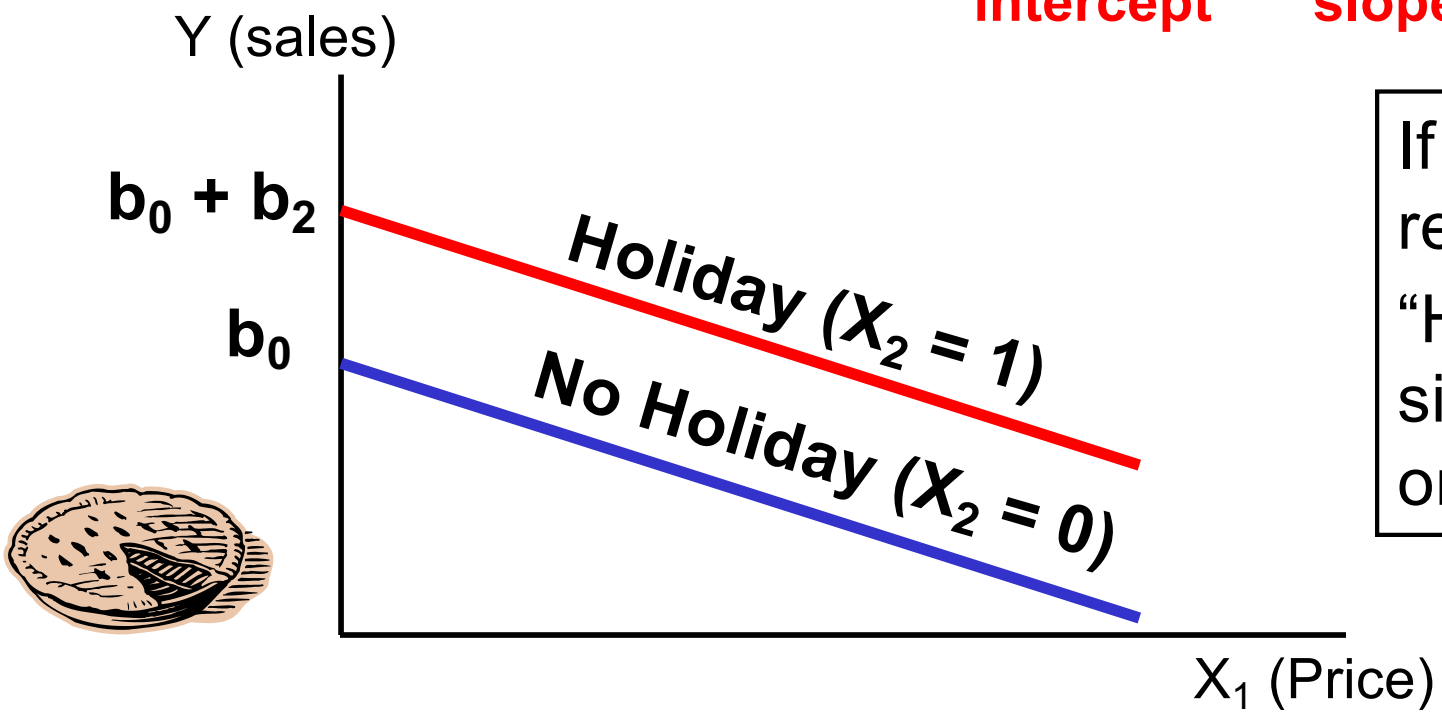
X_2 = holiday (X_2 = 1 if a holiday occurred during the week)
(X_2 = 0 if there was no holiday that week)

Dummy-Variable Example (with 2 Levels)

(continued)

$\hat{Y} = b_0 + b_1X_1 + b_2(1) = (b_0 + b_2) + b_1X_1$	**Holiday**
$\hat{Y} = b_0 + b_1X_1 + b_2(0) = b_0 + b_1X_1$	**No Holiday**

Different intercept **Same slope**



If H_0: $\beta_2 = 0$ is rejected, then "Holiday" has a significant effect on pie sales

Interpreting the Dummy Variable Coefficient (with 2 Levels)

Example: $\widehat{\text{Sales}} = 300 - 30(\text{Price}) + 15(\text{Holiday})$

Sales: number of pies sold per week
Price: pie price in $

$$\text{Holiday:} \begin{cases} 1 & \text{If a holiday occurred during the week} \\ 0 & \text{If no holiday occurred} \end{cases}$$

b_2 = 15: on average, sales were 15 pies greater in weeks with a holiday than in weeks without a holiday, given the same price

Dummy-Variable Models (more than 2 Levels)

- The number of dummy variables is **one less than the number of levels**
- Example:

  Y = house price ; X_1 = square feet

- If style of the house is also thought to matter:

  Style = ranch, split level, condo

Three levels, so two dummy variables are needed

Dummy-Variable Models (more than 2 Levels)

(continued)

- Example: Let "condo" be the default category, and let X_2 and X_3 be used for the other two categories:

Y = house price

X_1 = square feet

X_2 = 1 if ranch, 0 otherwise

X_3 = 1 if split level, 0 otherwise

The multiple regression equation is:

$$\hat{Y} = b_0 + b_1X_1 + b_2X_2 + b_3X_3$$

Interpreting the Dummy Variable Coefficients (with 3 Levels)

Consider the regression equation:

$$\hat{Y} = 20.43 + 0.045X_1 + 23.53X_2 + 18.84X_3$$

For a condo: $X_2 = X_3 = 0$

$$\hat{Y} = 20.43 + 0.045X_1$$

For a ranch: $X_2 = 1$; $X_3 = 0$

$$\hat{Y} = 20.43 + 0.045X_1 + 23.53$$

With the same square feet, a ranch will have an estimated average price of 23.53 thousand dollars more than a condo

For a split level: $X_2 = 0$; $X_3 = 1$

$$\hat{Y} = 20.43 + 0.045X_1 + 18.84$$

With the same square feet, a split-level will have an estimated average price of 18.84 thousand dollars more than a condo.

Interaction Between Independent Variables

- Hypothesizes interaction between pairs of X variables
  - Response to one X variable may vary at different levels of another X variable

- Contains two-way cross product terms
  - $$\hat{Y} = b_0 + b_1X_1 + b_2X_2 + b_3X_3$$
    $$= b_0 + b_1X_1 + b_2X_2 + b_3(X_1X_2)$$

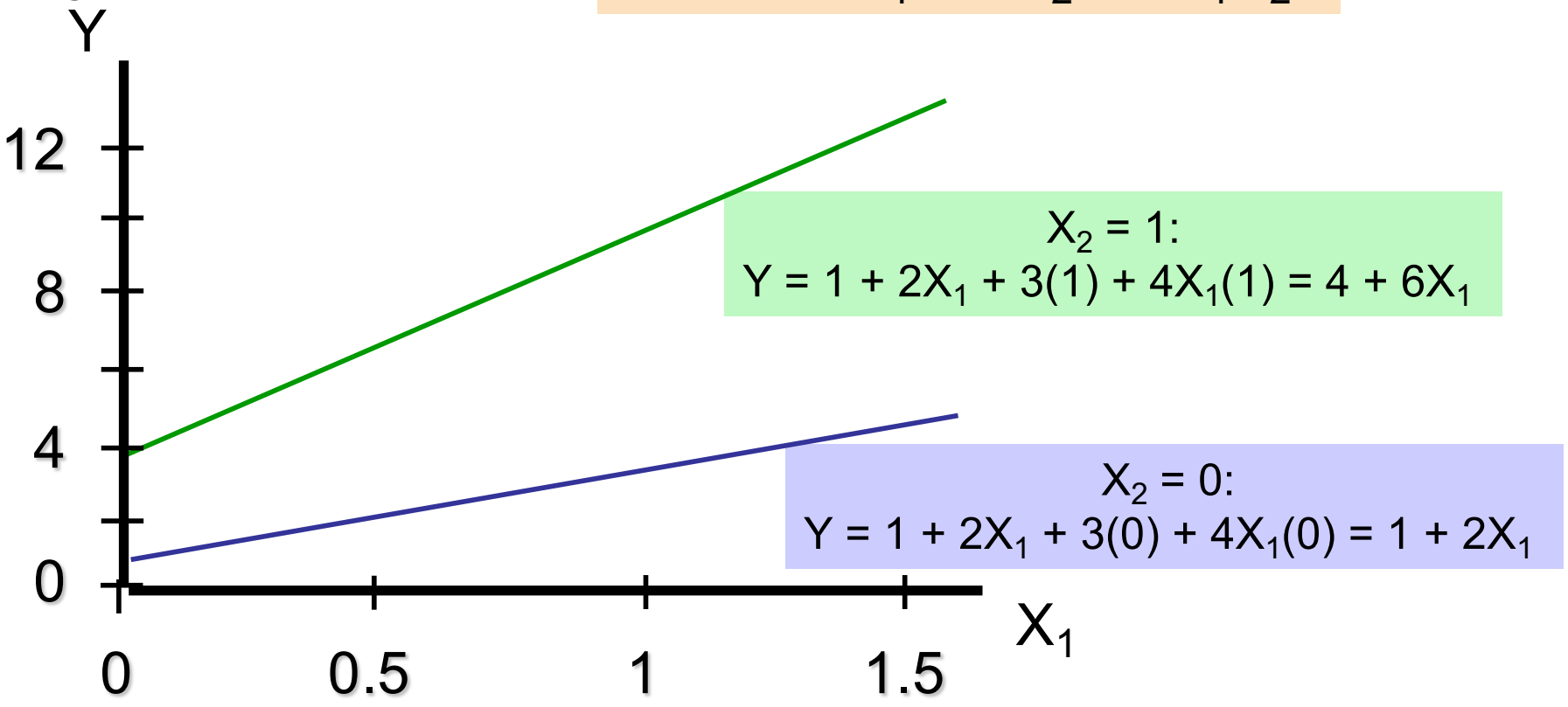
Effect of Interaction

- Given: $$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_1 X_2 + \varepsilon$$

- Without interaction term, effect of X_1 on Y is measured by β_1
- With interaction term, effect of X_1 on Y is measured by $\beta_1 + \beta_3 X_2$
- Effect changes as X_2 changes

Interaction Example

Suppose X_2 is a dummy variable and the estimated regression equation is $\hat{Y} = 1 + 2X_1 + 3X_2 + 4X_1X_2$



Slopes are different if the effect of X_1 on Y depends on X_2 value

Significance of Interaction Term

- Can perform a partial F test for the contribution of a variable to see if the addition of an interaction term improves the model

- Multiple interaction terms can be included
  - Use a partial F test for the simultaneous contribution of multiple variables to the model

Simultaneous Contribution of Independent Variables

- Use partial F test for the simultaneous contribution of multiple variables to the model
  - Let m variables be an additional set of variables added simultaneously
  - To test the hypothesis that the set of m variables improves the model:

$$F = \frac{[\text{SSR(all)} - \text{SSR (all except new set of m variables)}] / m}{\text{MSE(all)}}$$

(where F has m and n-k-1 d.f.)

Logistic Regression

- Used when the dependent variable Y is binary (i.e., Y takes on only two values)
- Examples
  - Customer prefers Brand A or Brand B
  - Employee chooses to work full-time or part-time
  - Loan is delinquent or is not delinquent
  - Person voted in last election or did not
- Logistic regression allows you to predict the probability of a particular categorical response

Logistic Regression

(continued)

- Logistic regression is based on the **odds ratio**, which represents the probability of a success compared with the probability of failure

$$\text{Odds ratio} = \frac{\text{probability of success}}{1 - \text{probability of success}}$$

- The logistic regression model is based on the natural log of this odds ratio

Logistic Regression

(continued)

Logistic Regression Model:

$$\ln(\text{odds ratio}) = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \ldots + \beta_k X_{ki} + \varepsilon_i$$

Where k = number of independent variables in the model

ε_i = random error in observation i

Logistic Regression Equation:

$$\ln(\text{estimated odds ratio}) = b_0 + b_1 X_{1i} + b_2 X_{2i} + \ldots + b_k X_{ki}$$

Estimated Odds Ratio and Probability of Success

- Once you have the logistic regression equation, compute the estimated odds ratio:

$$\text{Estimated odds ratio} = e^{\ln(\text{estimated odds ratio})}$$

- The estimated probability of success is

$$\text{Estimated probability of success} = \frac{\text{estimated odds ratio}}{1 + \text{estimated odds ratio}}$$

Chapter Summary

- Developed the multiple regression model
- Tested the significance of the multiple regression model
- Discussed adjusted r^2
- Discussed using residual plots to check model assumptions
- Tested individual regression coefficients
- Tested portions of the regression model
- Used dummy variables
- Evaluated interaction effects
- Discussed logistic regression